Risk-Adjusted Momentum (RAM)

A Quantitative Framework for Conditional Crypto Exposure

Michael Hyland February 2026

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1. Introduction

Bitcoin (BTC) has delivered strong long-term returns but with substantial volatility and repeated deep drawdowns. These characteristics make passive buy-and-hold exposure difficult to maintain as a strategic allocation.

This paper introduces **Risk-Adjusted Momentum (RAM)** - a systematic framework designed to condition crypto exposure on recent return per unit of risk. Rather than maintaining constant market exposure, RAM dynamically allocates capital across large-cap cryptocurrencies only when statistical conditions meet predefined thresholds.

The central research question is:

> Can conditional exposure based on risk-adjusted momentum improve the risk efficiency of crypto investing relative to passive BTC buy-and-hold?

The analysis uses daily data from January 2021 through December 2025 and explicitly separates parameter selection from out-of-sample evaluation to mitigate overfitting risk.

The objective is not to maximize historical returns, but to demonstrate a disciplined research process for developing and validating a conditional allocation framework in a highly volatile asset class.

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2. Research Framework

This study follows a structured quantitative research protocol designed to minimize overfitting and maximize generalizability.

The process consists of:

1. Defining an economic objective
2. Formulating explicit, testable hypotheses
3. Constructing a systematic signal using observable data
4. Selecting model parameters within a predefined training period
5. Evaluating robustness across multiple performance metrics
6. Validating results on an independent out-of-sample dataset

Parameter selection was completed using training data only. No model specifications were altered after observing out-of-sample results.

This separation between calibration and validation is critical. The objective of the framework is not to maximize in-sample performance, but to identify a stable allocation rule capable of persisting across different market regimes.

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3. Hypotheses

The RAM framework is designed to test two primary hypotheses.

H1: Risk-Adjusted Momentum Persistence

> Cryptocurrencies exhibiting high recent return per unit of risk will, on average, generate superior near-term performance relative to assets with weaker signals.
>
> Unlike traditional momentum measures based on absolute returns, RAM normalizes returns by recent volatility. This formulation tests whether momentum persistence remains statistically meaningful when adjusted for risk.
>
> Formally, the hypothesis implies that ranking assets by recent Sharpe-like measures should produce positive forward excess returns relative to lower-ranked assets.

H2: Cross-Asset Diversification Improves Risk Efficiency

> Allocating capital across multiple large-cap cryptocurrencies reduces overall portfolio volatility and drawdown due to imperfect cross-asset correlation.
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> This hypothesis evaluates whether diversification across correlated, but not perfectly correlated, crypto assets enhances risk-adjusted performance relative to concentrated BTC exposure.

Together, these hypotheses suggest that a dynamically allocated, multi-asset framework may deliver superior risk-adjusted performance relative to passive BTC exposure.

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4. Strategy Overview

Investment Universe

> The strategy is applied to the five largest non-stable cryptocurrencies by market capitalization:
>
> - BTC

- ETH
- XRP
- SOL
- BNB

Limiting the universe to large-cap assets reduces liquidity and implementation risk while preserving exposure to the dominant segments of the crypto market.

Signal Definition

For each asset i, a rolling risk-adjusted momentum measure is computed over an observation window of length n:

$$RAM_{i,t} = \frac{R_{i,t-n:t}}{\sigma_{i,t-n:t}}$$

where:

- $R_{i,t-n:t}$ is cumulative return over the prior n days
- $\sigma_{i,t-n:t}$ is the corresponding realized volatility

This statistic measures recent return per unit of risk.

Trading Logic

An asset is held long if the magnitude of its RAM signal exceeds a predefined threshold $\pm ST$.

Importantly:

- Positive threshold breaches capture traditional momentum persistence.
- Negative threshold breaches capture volatility-induced reversal effects observed during exploratory analysis.

No short positions are taken. The strategy is long-only or neutral.

Capital Allocation

Capital is allocated equally across each asset (20% of total capital for each cryptocurrency). The amount allocated to each asset is invested on an all-or-none basis.

Uninvested capital remains in cash (no yield assumed).

The strategy does not employ leverage.

All performance statistics are calculated on total capital, including periods held in cash.

Rebalancing

Signals are computed using closing prices.

Positions are adjusted daily using only information available at the close of day t, with trades executed at the next session.

This design eliminates look-ahead bias and ensures implementability.

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5. Execution Assumptions

Signals are computed using daily closing prices and therefore become available only after the close of day t.

Trades dictated by the signal are executed at the next available session.

Given the continuous 24-hour trading structure of major cryptocurrencies, the next session open is assumed to occur at or near the prior closing price. For modeling purposes, next-day open prices are approximated using prior-day closing prices.

This assumption reflects the minimal overnight gap risk typical of continuously traded crypto assets but should be re-evaluated prior to live deployment.

Transaction costs of 0.16 basis points per trade are applied. This reflects exchange-level trading fees for large-cap crypto pairs under standard liquidity conditions.

No additional slippage is modeled. Given the liquidity profile of the selected assets, slippage is assumed to be modest relative to the magnitude of the signal threshold. However, real-world implementation would require dynamic slippage estimation and monitoring.

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6. Data and Implementation Assumptions

Universe Construction

The investment universe consists of the five largest non-stable cryptocurrencies by current market capitalization.

This static universe selection introduces survivorship bias, as assets are chosen based on present-day prominence rather than historical eligibility. A more rigorous

implementation would dynamically adjust the universe through time based on rolling market capitalization rankings.

The current approach was adopted for tractability and to focus on liquidity-stable assets.

Data Source

Daily closing prices were obtained from Binance for the period January 2021 through December 2025.

All signals are computed using only prior information. No forward-looking data is incorporated.

Calibration and Evaluation Periods

- **Training (Calibration) Period:** January 2021 – December 2023
- **Testing (Out-of-Sample) Period:** January 2024 – December 2025

The training period includes both high-volatility bull and bear regimes. The testing period includes a market peak in October 2024 followed by reversal, providing exposure to multiple volatility environments.

Model parameters were selected exclusively using training data. No parameter adjustments were made after observing out-of-sample results.

This separation preserves the integrity of the validation process and mitigates overfitting risk.

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7. Surprising Signal Behavior

Initial hypothesis testing focused on positive RAM threshold breaches, consistent with traditional momentum persistence.

As a natural extension, symmetric testing was conducted for negative RAM threshold breaches. The original expectation was that extreme negative risk-adjusted momentum would justify reduced exposure.

However, empirical results indicated the opposite effect.

Extreme negative RAM values were associated with short-term return reversals, consistent with volatility-driven overreaction and forced liquidation dynamics commonly observed in crypto markets.

This behavior was observed during the training phase and subsequently validated in the out-of-sample period without modification to model parameters.

The evidence suggests that both:

- Positive RAM extremes (trend continuation), and
- Negative RAM extremes (volatility-induced mean reversion)

carry predictive information.

Incorporating both threshold types into the allocation rule materially improved overall risk-adjusted performance.

A detailed breakdown of the separate positive and negative signal analyses is provided in Appendix I.

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8. Model Selection and Robustness

Model performance was evaluated jointly using:

- Sharpe ratio
- Maximum drawdown

These metrics balance return efficiency with capital preservation.

Two parameters govern the RAM signal:

- Observation window length n
- Signal threshold $\pm ST$

Parameter ranges were intentionally constrained to mitigate multiple-testing bias:

- n: 5–20 days
- ST: 0.10 to 0.40 (with limited extension to 0.50 for shorter windows)

Rather than selecting a single parameter combination that maximized in-sample Sharpe ratio, candidate parameters were required to lie within a broad region of stable outperformance across neighboring values.

This stability criterion reduces the likelihood that results are driven by isolated local maxima and increases the probability of generalization.

Because the strategy is invested intermittently, the effective sample size is smaller than the total number of daily observations. Accordingly, robustness was assessed across multiple performance metrics rather than relying solely on point estimates.

To control model complexity, two simplifying constraints were imposed:

- All cryptocurrencies share identical observation windows and thresholds.
- Positive and negative thresholds are symmetric.

These constraints reduce parameter dimensionality and limit overfitting risk. Future research may relax these assumptions if stability persists under stricter validation.

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9. In-Sample Results (2021–2023)

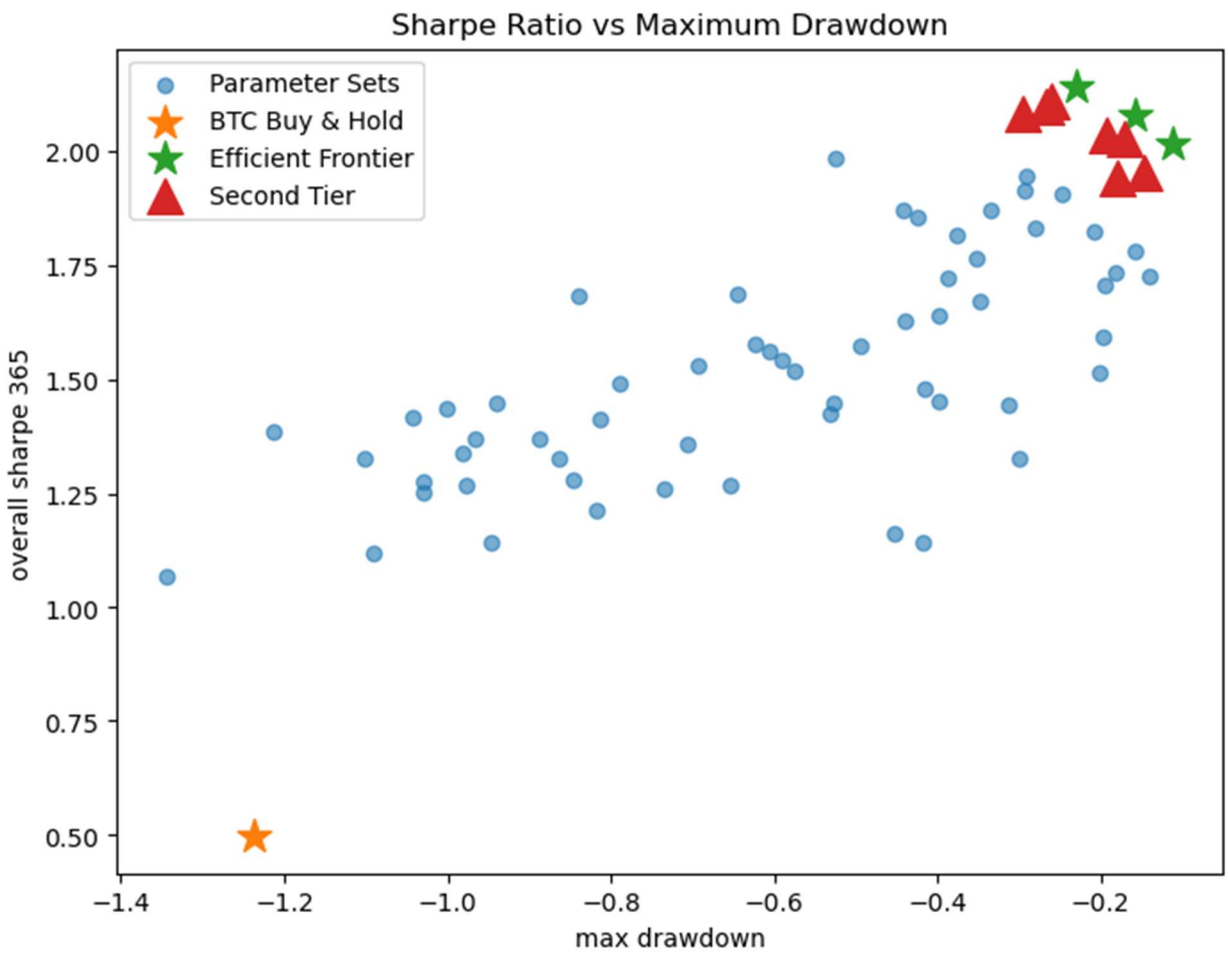
Appendix I provides a detailed account of the training-period development process.

To control model complexity and limit multiple-testing bias, parameter exploration was intentionally constrained to a predefined grid:

- **Observation window (n):** 5 to 20 trading days
- **Signal threshold (±ST):** 0.10 to 0.40 in 10 basis point increments, with limited extension to 0.50 for shorter observation windows (5–10 days)

This grid produced 69 parameter combinations for evaluation.

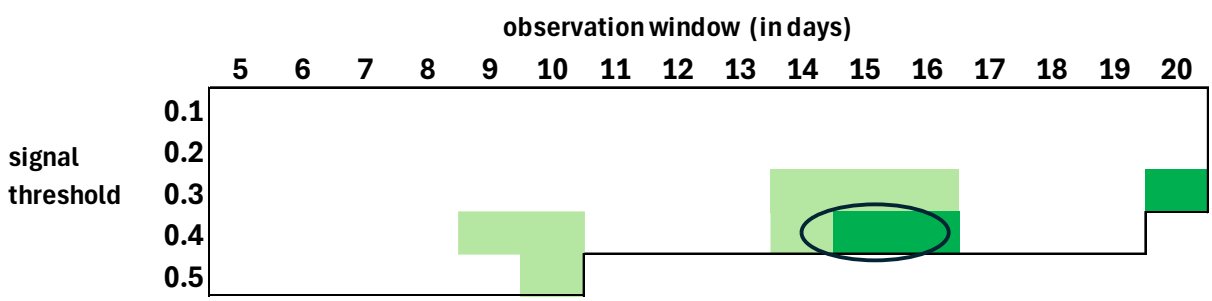
The figure below presents the Sharpe ratio versus maximum drawdown for each combination during the training period, with BTC buy-and-hold (BTC BAH) shown for reference.



All but one of the parameter combinations outperformed BTC BAH on a risk-adjusted basis during the calibration period. Importantly, performance was not concentrated in a single isolated peak; instead, a broad region of parameter combinations exhibited similar Sharpe and drawdown characteristics.

Consistent with the evaluation framework described earlier, parameter selection prioritized stability across neighboring values rather than maximizing a single in-sample Sharpe ratio.

The heatmap below highlights the frontier region (bright green) and second-tier combinations (lighter green), illustrating a contiguous area of robust performance.



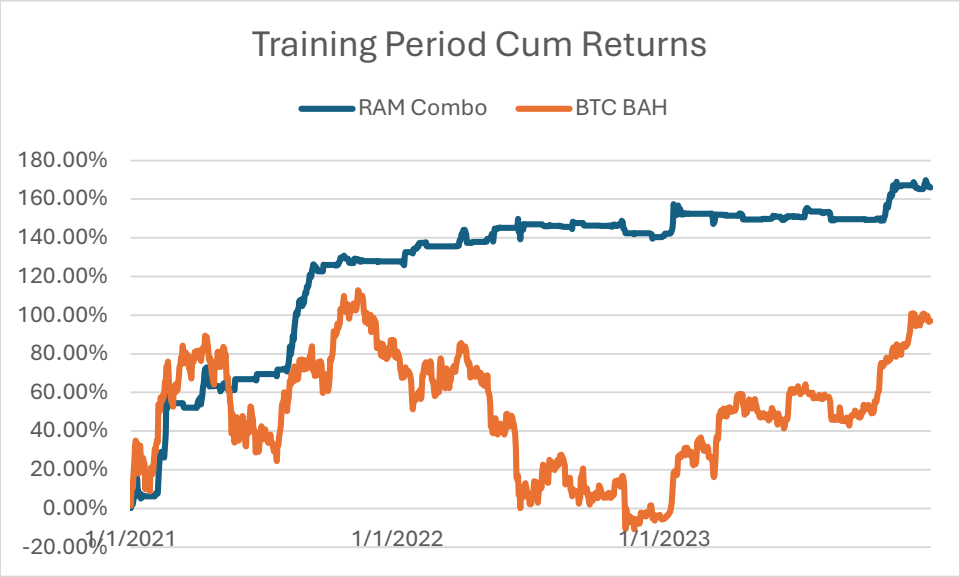
Within this stability region, the parameter sets $[15, \pm 0.4]$and $[16, \pm 0.4]$displayed nearly identical performance across all primary metrics. Rather than selecting a single specification, a 50/50 combination of the two was constructed to reduce parameter sensitivity and improve robustness.

Training-period performance is summarized below:

Statistic	[15, +/-0.4]	[16, +/-0.4]	Combo	BTC BAH
CAGR	75%	72%	75%	38%
Max DD	-16%	-11%	-16%	-124%[1]
Sharpe	2.08	2.02	2.10	0.50

The combination portfolio achieved a meaningfully higher Sharpe ratio and substantially reduced drawdowns relative to BTC buy-and-hold.

The equity curve comparison below illustrates the smoother return profile of the RAM framework during the training period.



The combination portfolio exhibited a 35% correlation with BTC during the calibration window, indicating that performance improvements were not solely attributable to reduced exposure but also to conditional allocation and cross-asset diversification effects.

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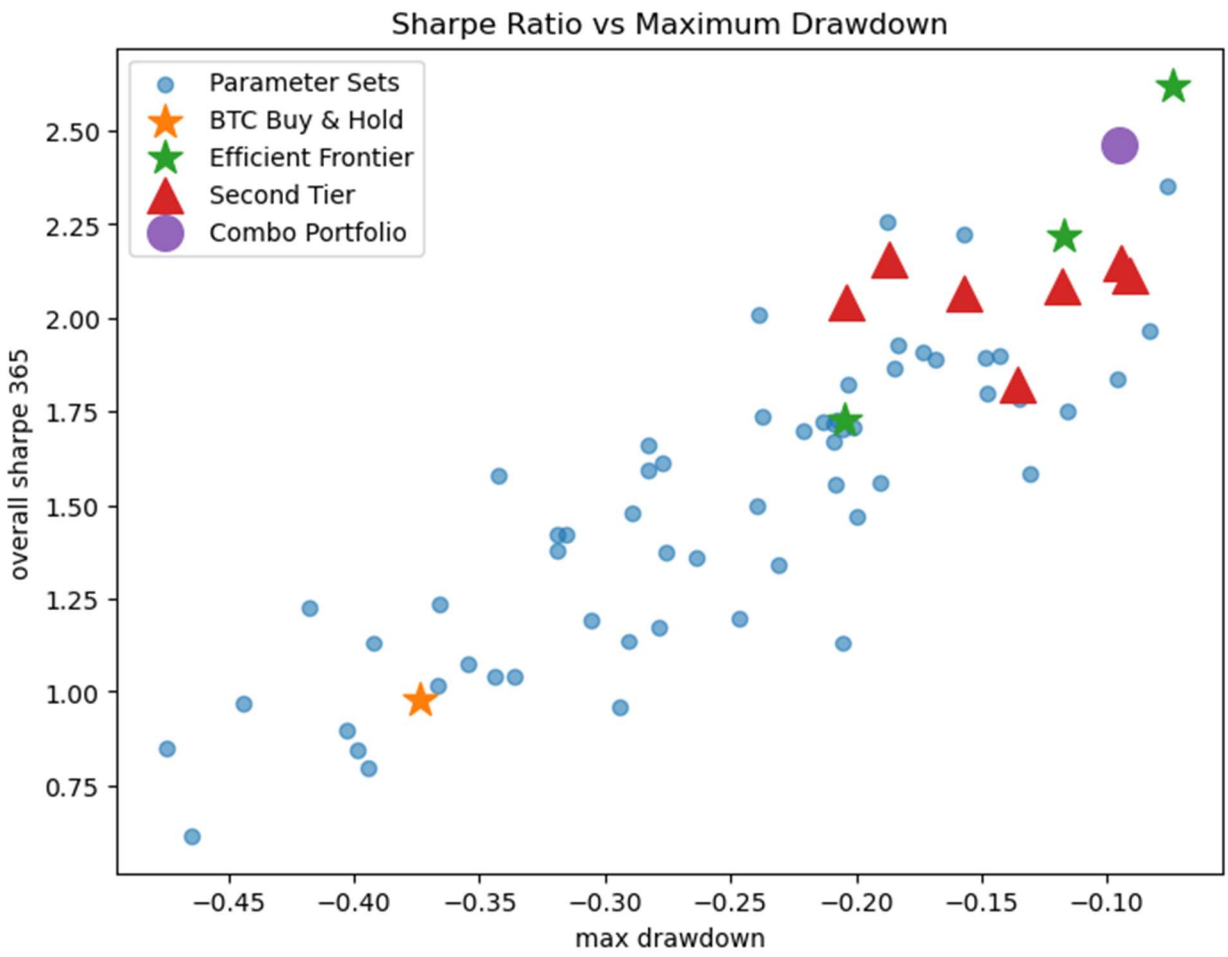
[1] Obviously, the real compounded maximum drawdown cannot be more than 100%. But, the maximum drawdown statistic shown is the based on the simple sum of daily percentage returns and not the compounded daily percentage returns.

10. Out-of-Sample Validation (2024–2025)

All model parameters, signal definitions, and allocation rules were fixed at the conclusion of the training period. No adjustments were made after observing out-of-sample results.

The figure below presents the Sharpe ratio versus maximum drawdown for the same 69 parameter combinations during the testing period. Parameter sets identified as top-tier during the training phase retain their markings for reference.



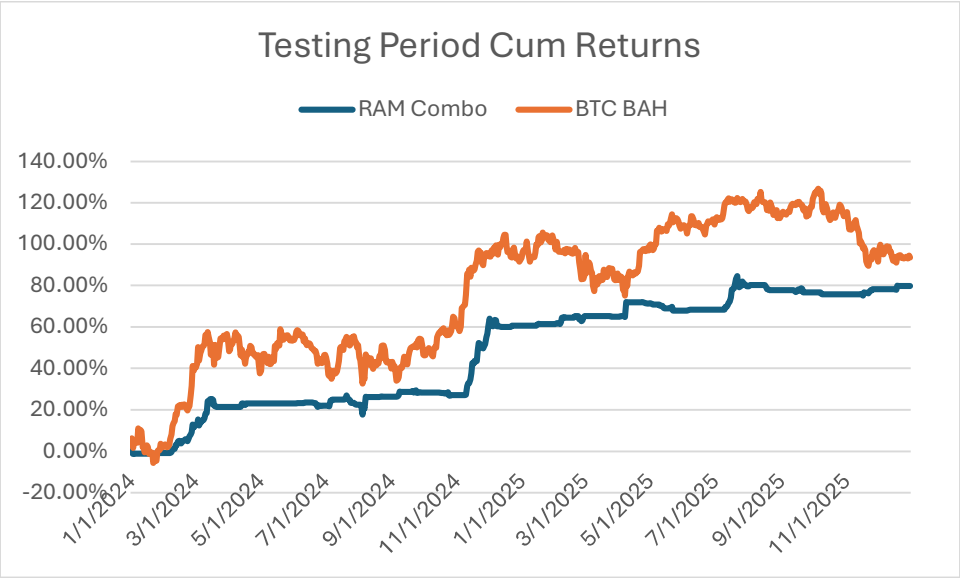
Several observations emerge:

- A meaningful subset of parameter combinations continued to outperform BTC buy-and-hold on a risk-adjusted basis.
- Parameter sets identified within the training-period stability region remained competitive relative to the full grid.
- The 50/50 combination portfolio maintained strong risk-adjusted performance.

Unlike the training period, BTC buy-and-hold generated a higher absolute return during the testing window. However, the RAM framework continued to deliver materially lower volatility and reduced drawdowns.

Out-of-sample performance is summarized below:

Statistic	[15, +/-0.4]	[16, +/-0.4]	Combo	BTC BAH
CAGR	46%	52%	49%	59%
Max DD	-12%	-7%	-10%	-37%
Sharpe	2.22	2.62	2.46	0.98



Although the combination portfolio underperformed BTC in absolute return during this period, it achieved significantly higher Sharpe ratios and substantially lower maximum drawdowns.

The correlation between the combination portfolio and BTC during the testing period was 39%, consistent with the diversification and conditional allocation characteristics observed in-sample.

Further analysis provides insight into how returns were generated.

Over the two-year testing period:

- The strategy remained uninvested approximately 68% of the time.
- Full exposure to all five assets occurred on only nine trading days.

This confirms that performance was driven by selective regime participation rather than continuous market exposure.

# of Investments	# of Days	% of Days
0	494	68%
1	100	14%
2	58	8%
3	48	7%
4	22	3%
5	9	1%
sum	731	100%

Capital allocation across individual assets was broadly distributed, with no single cryptocurrency dominating signal activation.

Crypto	# of Days	% of Days
BTC	106	15%
ETH	120	16%
BNB	95	13%
XRP	95	13%
SOL	77	11%

Additionally, both positive and negative threshold breaches contributed meaningfully to total returns:

Signal Threshold Breach	% of Returns
positive	67%
negative	33%
sum	100%

This balanced contribution supports the dual-signal design and reinforces the robustness of the framework across different market conditions.

The out-of-sample results demonstrate that:

- Risk-adjusted improvements observed during calibration were not entirely dependent on in-sample conditions.
- The selected parameter region generalized reasonably well across a distinct market regime.
- The primary benefit of the RAM framework lies in volatility and drawdown mitigation rather than absolute return maximization.

Importantly, no parameter re-optimization was performed during the testing period.

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11. Economic Interpretation

The empirical results align with several structural characteristics of crypto markets.

Momentum Persistence

Crypto markets are heavily influenced by:

- Retail participation
- Leveraged positioning
- Perpetual futures funding dynamics
- Trend-following behavior

These forces contribute to short- and medium-term momentum persistence, particularly during stable trending regimes. Normalizing returns by recent volatility helps identify periods in which price moves are directional rather than noise-driven.

This supports Hypothesis 1: risk-adjusted momentum contains forward-looking information.

Volatility Shocks and Reversal Dynamics

Crypto markets are prone to abrupt volatility spikes driven by liquidations, funding imbalances, and sentiment shifts.

Extreme negative RAM readings often coincide with forced deleveraging events. These episodes can produce short-term overshooting followed by mechanical mean reversion as liquidity stabilizes.

The inclusion of negative threshold signals captures this dynamic.

Diversification and Imperfect Correlation

Although major cryptocurrencies are positively correlated, correlations are far from perfect and vary across regimes.

Cross-asset diversification reduces concentration risk relative to BTC-only exposure. This supports Hypothesis 2 and contributes materially to the reduction in drawdown observed during both the training and testing periods.

Conditional Exposure and Regime Filtering

Unconditional crypto exposure subjects capital to high-volatility regime transitions, which historically account for a disproportionate share of drawdowns.

By conditioning exposure on statistically favorable environments, RAM reduces participation during unstable periods while remaining invested during persistent trends.

This regime-filtering mechanism explains the improved risk-adjusted profile relative to passive BTC buy-and-hold.

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12. Limitations and Risks

While results are encouraging, several structural and implementation risks should be acknowledged.

Structural Market Risk

The strategy remains fully exposed to crypto market dynamics when invested. A prolonged structural shift - such as reduced volatility, diminished momentum persistence, or tighter cross-asset correlation - could impair signal effectiveness.

As crypto markets mature and institutional participation increases, historical behavioral patterns may weaken.

Regime Sensitivity

Performance is driven by episodic exposure rather than continuous trading. Because the strategy is invested intermittently, realized outcomes are sensitive to the timing of signal activation.

Extended periods without strong trends or repeated volatility spikes could reduce effectiveness.

Parameter Stability

Although parameter selection emphasized stability regions, model performance depends on the persistence of statistical relationships observed during the calibration period.

Continuous monitoring would be required to detect structural drift in signal behavior.

Execution Assumptions

Execution modeling assumes minimal overnight gap risk and limited slippage in large-cap assets.

Real-world implementation may experience:

- Wider spreads during stress periods
- Increased slippage
- Temporary liquidity dislocations

Transaction costs and execution quality should therefore be stress-tested prior to deployment.

Capacity Constraints

As strategy scale increases, impact costs may rise, particularly during volatility events when signals are most active.

Although the selected assets are among the most liquid in crypto markets, large-scale deployment could reduce realized Sharpe ratios.

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13. Conclusion

This study demonstrates a disciplined quantitative research process applied to conditional crypto allocation.

Beginning with a clearly defined economic objective - improving risk efficiency relative to passive BTC exposure - the RAM framework was constructed, parameterized using a predefined calibration window, and validated on an independent out-of-sample dataset without modification.

The strategy achieved:

- Comparable long-term returns to BTC buy-and-hold
- Materially lower volatility
- Significantly reduced drawdowns
- Reduced market exposure through conditional participation

Importantly, these results were obtained without post-hoc parameter adjustment or optimization based on out-of-sample performance.

The primary contribution of this paper is methodological rather than predictive. It illustrates how hypothesis-driven design, constrained parameter exploration, stability-region selection, and clean train/test separation can produce robust allocation frameworks in highly volatile asset classes.

Future research may explore asymmetric thresholds, asset-specific parameters, and alternative volatility normalizations. However, the core finding remains:

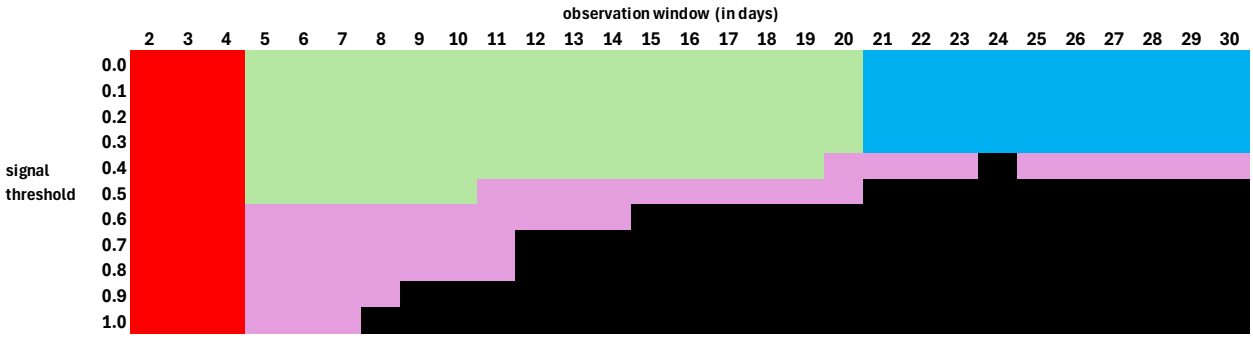
> Conditional exposure frameworks can materially improve the risk profile of crypto allocations while preserving upside participation.

Appendix – Training Period Process

Initial exploratory analysis focused on positive RAM threshold breaches, consistent with the primary momentum persistence hypothesis.

A structured grid search was conducted across a wide range of parameter. The resulting performance landscape was categorized into regions based on signal frequency, stability, and robustness:

- **Red:** Unstable results driven by very short observation windows
- **Black:** No activation due to excessively high thresholds
- **Purple:** Low trade frequency leading to unstable Sharpe estimates
- **Blue:** Extended parameter region reserved for future exploration
- **Green:** Stable region of consistent outperformance



The green region formed the focus of subsequent analysis.

Following evaluation of positive threshold behavior, symmetric testing was conducted for negative threshold breaches.

The initial expectation was that extreme negative RAM readings would justify reduced exposure. However, systematic testing revealed that negative extremes were frequently followed by short-term reversals.

This behavior is consistent with volatility-induced overreaction and forced liquidation dynamics.

Importantly, this effect was identified during the training phase and subsequently validated out-of-sample without modifying parameter ranges.

The complementary nature of positive-momentum persistence and negative-reversal effects suggested a unified framework incorporating both signals.

The signal threshold at zero was excluded after preliminary testing demonstrated that it resulted in continuous market exposure, eliminating the conditional allocation hypothesis being evaluated.

Removing the zero-threshold ensured that the framework remained a regime-filtering strategy rather than a fully invested momentum portfolio.

To prevent combinatorial explosion and excessive data-mining risk, model dimensionality was intentionally constrained:

- All assets share identical parameter values
- Positive and negative thresholds remain symmetric
- Observation windows are uniform across assets

In theory, the model could allow:

- Asset-specific thresholds
- Asymmetric positive/negative levels
- Asset-specific observation windows
- Dynamic universe selection

These extensions were intentionally excluded to limit parameter dimensionality and preserve statistical integrity during initial development.

Future research may explore these extensions subject to robust validation procedures.